

ECSE 2500

Lec 3

Jan. 18th

Topic: Axioms of probability

Rules

Last lecture, we have introduced several key elements of random experiments such as sample space, events and event class.

Ex. Flip coin twice, record the results of two flips

Sample space $S := \{HH, HT, TH, TT\}$

Event class (all possible subsets of S)

$$F := \left\{ \begin{array}{l} \emptyset, \{HH\}, \{HH, HT\}, \{HH, HT, TH\}, \{HH, HT, TH, TT\} \\ \{HT\}, \{HH, TH\}, \{HH, HT, TT\} \\ \{TH\}, \{HH, TT\}, \{HH, TH, TT\} \\ \{TT\}, \{HT, TH\}, \{HT, TH, TT\} \\ \{HT, TT\} \\ \{TH, TT\} \end{array} \right\}$$

$$1 + 4 + 6 + 4 + 1 = 16 \\ = 2^4$$

Now we want to assign a probability to each event A in the event class F , denoted $P(A)$.

$$P(\cdot) : \text{Event } A \rightarrow \text{Scalar value } [0,1]$$

Function: A mapping between input $\rightarrow$ output

In order to be self-consistent, the probability function $P(\cdot)$ must satisfy 3 Axioms/Rules:

Foundation
of discrete
probability

1. $P(A) \geq 0$ (No negative probability)
2. $P(S) = 1$ (There is a fixed "amount" or "mass" of probability = 1)
3. If $A \cap B = \emptyset$ then

$$P(A \cup B) = P(A) + P(B)$$

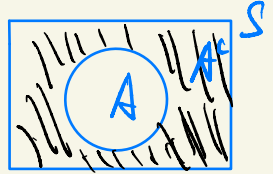
Remark: (3) If events are disjoint (no overlap in common), probability of union is the sum of individual events.

Technically, we can generalize to 3 or more events
If $A_1, A_2, \dots, A_n$ is a set of events with $A_i \cap A_j = \emptyset, \forall i, j$, then $P(\bigcup_{i=1}^n A_i) = \sum_{i=1}^n P(A_i)$

→ From these basic principles, we can derive several other important properties of probability.

$$1. P(A^c) = 1 - P(A)$$

Complement of A



Since we have $P(A \cup A^c) = P(S) \stackrel{\text{Axiom 2}}{=} 1$
 $\stackrel{\text{Axiom 3}}{=} P(A) + P(A^c)$

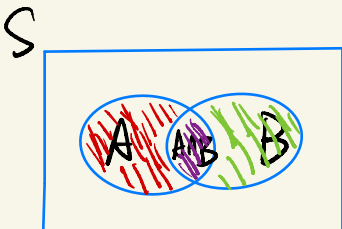
$$\Rightarrow P(A^c) = 1 - P(A) \quad \text{Q.E.D.}$$

2. $P(A) \leq 1$

since $P(A) = 1 - \underbrace{P(A^c)}_{\geq 0} \stackrel{\text{Axiom 1}}{\leq} 1$

3. $P(\phi) = 0$, since $P(\phi) = 1 - P(S) \stackrel{\text{Axiom 2}}{=} 0$

4. $P(A \cup B) = P(A) + P(B) - P(A \cap B)$



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$$A \cup B = (A-B) \cup (A \cap B) \cup (B-A)$$

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Disjoint sets

Since $P(A \cup B) \stackrel{\text{Axiom 3}}{=} P(A-B) + P(A \cap B) + P(B-A)$

$$= \underbrace{P(A-B) + P(A \cap B)}_{\text{Axiom 3}} + \underbrace{P(B-A) + P(A \cap B)}_{\text{Axiom 3}} - P(A \cap B)$$

$$\stackrel{\text{Axiom 3}}{=} P(A) + P(B) - P(A \cap B), \text{ Q.E.D.}$$

5. If $A \subset B$, $P(A) \leq P(B)$

Since $P(B) \stackrel{\text{Axiom 3}}{=} P(A) + \underbrace{P(B-A)}_{\text{Disjoint sets}}$

$$\stackrel{\text{Axiom 1}}{\geq} P(A)$$

□ Discrete, Finite Sample Spaces

If we have a finite number of outcomes, then any event is just a subset of outcomes, and the probability is the sum of the individual probabilities.

Exp. Roll a 6-sided die and observe the number

$$S := \{1, 2, 3, 4, 5, 6\}$$

$$\text{Event } A := \{\text{Number is even}\} = \{2, 4, 6\}$$

$$P(\text{Number is even}) = P(\{2, 4, 6\})$$

$$= P(\{2\} \cup \{4\} \cup \{6\})$$

$$\stackrel{\text{Axiom 3}}{=} P(\{2\}) + P(\{4\}) + P(\{6\})$$

$$= \frac{1}{6} + \frac{1}{6} + \frac{1}{6} = \frac{1}{2}$$

This also works for discrete sample space with an infinite number of outcomes.

Exp. Roll a fair 6-sided die, until the current top number matches the previous top number, and record the number of rolls as X .

In this experiment, $S := \{2, 3, 4, \dots\}$

However, we can still compute the probability of rolling a certain number of times.

$$P(X=1) = 0$$

$$P(X=2) = \frac{1}{6} \quad \text{How?}$$

6 outcomes that have same number twice
36 total outcomes

$P(X=3)$ is a little more complicated

We need Roll 3 = Roll 2, but also

Roll 2 $\neq$ Roll 1

$$P(X=3) = \left(1 - \frac{\overset{\downarrow}{6 \text{ outcomes}}}{36 \text{ outcomes}}\right) \times \left(\frac{6 \text{ outcomes}}{36 \text{ outcomes}}\right)$$

$$= \frac{5}{6} \times \frac{1}{6} = \frac{5}{36} \approx 0.13$$

In general, we can show that

$$P(X=n) = P(\text{roll } 2 \neq \text{roll } 1)$$

$$\bullet P(\text{roll } 3 \neq \text{roll } 2)$$

$$\bullet P(\text{roll } 4 \neq \text{roll } 3)$$

$$\bullet P(\text{roll } n = \text{roll } n-1)$$

$$= \left(\frac{5}{6}\right)^{n-2} \left(\frac{1}{6}\right) \quad \text{works } n=2, 3 \text{ as well}$$